

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

This section focuses on articulating how your proposed solution is supported by STEM (science, technology, engineering, mathematics, computer science, etc.) concepts. Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice. Go beyond describing the engineering design process to describe the STEM concepts as they apply to your proposed solution in order to meet the design requirements. Be sure to apply the applicable STEM concepts in context, directly to the specifics of your design. Review concepts learned in your other classes and tackle new concepts as appropriate. Include any relevant calculations.

C3 indicates that you should have at least three high-priority design requirements to receive a score of 3 (and four for a score of 4, and five for a 5). Your number of design requirements and your number of high priority design requirements should agree throughout your portfolio. You should use the same five design requirements here too, and apply at least four of them to STEM concepts to achieve a score of 5 on E1. Use as many copies of this sub-template as needed.

STEM Principle for High-Priority Design Requirement #1 - **Drake B.**

STEM Concept: **“R.E.S.C.U.E.R.”**

Describe how STEM concept applies directly to the specifics of your design:

STEM applies to my design by utilizing Technology to construct the programming of the button in order for it to activate upon being pushed and send signals for tracking and alert authorities, Engineering being used to construct the casing of the buttons circuits to be protected and allow for attachment Points to be used when applying custom attachments for different button forms; while mathematics and science being very scarce in the process, they still aid in the measurements needed to allow the case to hold circuits and be optimal for hiding but usable

Science: 2/5

Technology: 5/5

Engineering: 5/5

Mathematics: 2/5

STEM Principle for High-Priority Design Requirement #2 - **Quetzali C.**

STEM Concept: **“C.S.P. R.”**

Describe how STEM concept applies directly to the specifics of your design:

STEM applies to my design by... My design doesn't really involve science since the whole point of this project mostly involves the technology and engineering aspect, science can't correlate with what we're doing. We had to program and use a micro bit for this project to use Technology. For Engineering we had to figure out a way to solve this problem using the resources available

to us and make a product that comes with this solution. We have to do measurements for how big the microchip is and make sure it's not big enough to wherever we're inserting it to use Mathematics.

Science: 1/5
Technology: 5/5
Engineering: 5/5
Mathematics: 3/5

STEM Principle for High-Priority Design Requirement #3 - **Bryson W.**

STEM Concept: **"Button"**

Describe how STEM concept applies directly to the specifics of your design:

STEM applies to my design: **Science- is used for the experimentation process .Technology- is used to make the microbit and make it more useful. Engineering- is used when designing the microbit to do what we want. Math- is used for the size needed for the kind of person the micro bit is attached to.**

Science: 2/5
Technology: 5/5
Engineering: 4/5
Mathematics: 2.5/5

STEM Principle for High-Priority Design Requirement #4 - **Lilly H.**

STEM Concept: **"C.T. D"**

Describe how STEM concept applies directly to the specifics of your design:

STEM applies to my design by... Using technology to design the product itself, engineering by thinking through ideas and planning how to design is going to work. Math is used by counting and making measurements during the procedure of designing and putting together the microbit. Even though not a lot, science has some important parts with experimenting the microbit.

Science: 2/5
Technology: 5/5
Engineering: 5/5
Mathematics: 3.5/5

STEM Principle for High-Priority Design Requirement #5 - **Kaylee O.**

STEM Concept: **"K.U.W. K"**

Describe how STEM concept applies directly to the specifics of your design:

STEM applies to my design by...

Science: 3/5

Technology: 5/5
Engineering: 5/5
Mathematics: 3/5

STEM Principle for High-Priority Design Requirement #6 - **Christian S.**

STEM Concept: "**C.A.N**"

Describe how STEM concept applies directly to the specifics of your design:

STEM applies to my design by using a lot of Technology and Engineering. But Mathematics and Science are lacking. Technology is the micro bit itself and the using of the micro bit to notify authorities and save the kid all from a little tech. Engineering is used in almost everything with the micro bit, from just coding it to putting it in the jewelry. Mathematics is the coding of the micro bit using math. Science is almost none, maybe the science is the wire connected from the micro bit to the phones of the authorities.

Science: 1/5
Technology: 5/5
Engineering: 5/5
Mathematics: 4/5

STEM Principle for High-Priority Design Requirement #7 - **Xavier B.**

STEM Concept: "**K.P.T**"

Describe how STEM concept applies directly to the specifics of your design:

STEM applies to my design by using the technology to track or give signals to where the object is engineering the watch will be on the risk of the person like a normal watch but it will have a tracker for the police and adults science still working on it

Science: 3/5
Technology: 5/5
Engineering: 5/5
Mathematics: 0/5

Experts-

Expert #1: Officer Obriant - The OPS school officer. .
Expert #2: School Teacher - The OPS teachers.
Expert #3: Students Parents - We didn't get any.

"Officer Obriant told us to change the main focus of the project. Change the tracking from children being kidnaped to just an all around tracker, which informed our decision to change/reconsider the main focus of our project".

"School Teacher told us a lot of general info about the project, which verified/reinforced our decision to keep the general focus of our project".

"Students' Parents told us NOTHING, which verified/reinforced our decision to NOT ASK PARENTS."